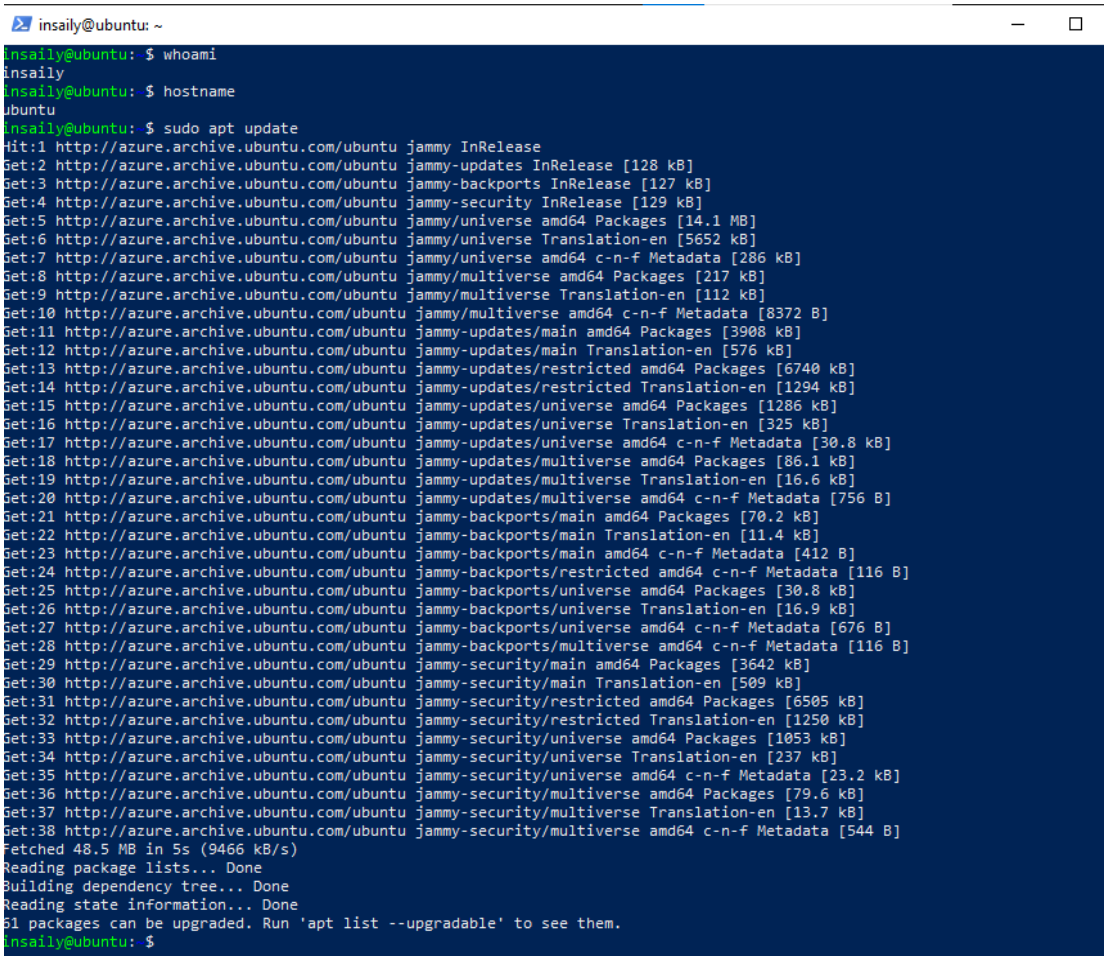


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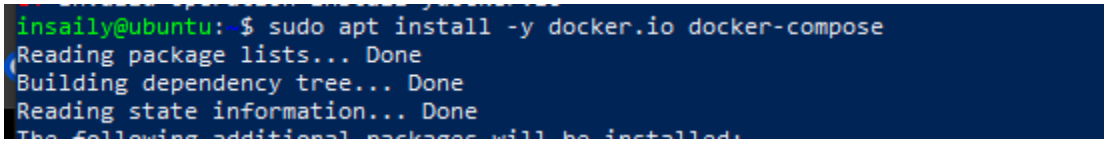
CLOUD COMPUTING
ASSIGNMENT #01

Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Connect to your assigned Ubuntu server.
package name is docker.io:



. Update the Ubuntu package index and install Docker Engine and curl. On Ubuntu, the required



Create the system-wide Docker CLI plug-in directory and download the Docker Compose plug-in for
an x86_64 Ubuntu server:

4. Start the Docker service and add your current Ubuntu account to the docker group:

```
Unpacking ubuntu-fan (0.12.16) ...
Setting up python3-dotenv (0.19.2-1) ...
Setting up python3-texttable (1.6.4-1) ...
Setting up python3-docopt (0.6.2-4) ...
Setting up dnsmasq-base (2.91-0ubuntu0.22.04.2) ...
Setting up runc (1.3.4-0ubuntu1~22.04.1) ...
Setting up dns-root-data (2024071801~ubuntu0.22.04.1) ...
Setting up bridge-utils (1.7-1ubuntu3) ...
Setting up pigz (2.6-1) ...
Setting up containerd (2.2.1-0ubuntu1~22.04.2) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /lib/systemd/system/containerd.service.
Setting up python3-websocket (1.2.3-1) ...
Setting up python3-dockerpty (0.4.1-2) ...
Setting up ubuntu-fan (0.12.16) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service → /lib/systemd/system/ubuntu-fan.service.
Setting up python3-docker (5.0.3-1) ...
Setting up docker.io (29.1.3-0ubuntu3~22.04.2) ...
Adding group `docker' (GID 122) ...
Done.
```

5. Verify the Git, Docker, and Docker Compose installations:

```
insaily@ubuntu:~$ sudo systemctl enable --now docker
insaily@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3~22.04.2
insaily@ubuntu:~$
```

```
insaily@ubuntu:~$ docker-compose --version
docker-compose version 1.29.2, build unknown
insaily@ubuntu:~$
```

```
insaily@ubuntu:~$ mkdir -p ~/gitea
insaily@ubuntu:~$ cd ~/gitea
insaily@ubuntu:~/gitea$ pwd
/home/insaily/gitea
insaily@ubuntu:~/gitea$
```

7. Use the existing compose.yaml without changing its server URL or port settings. Start Gitea and

```
insaily@ubuntu:~/gitea$ nano docker-compose.yml
insaily@ubuntu:~/gitea$ cat docker-compose.yml
services:
  db:
    image: postgres:15
    container_name: gitea-db
    restart: always
    environment:
      POSTGRES_USER: gitea
      POSTGRES_PASSWORD: gitea_db_pass_123
      POSTGRES_DB: gitea
    volumes:
      - postgres_data:/var/lib/postgresql/data

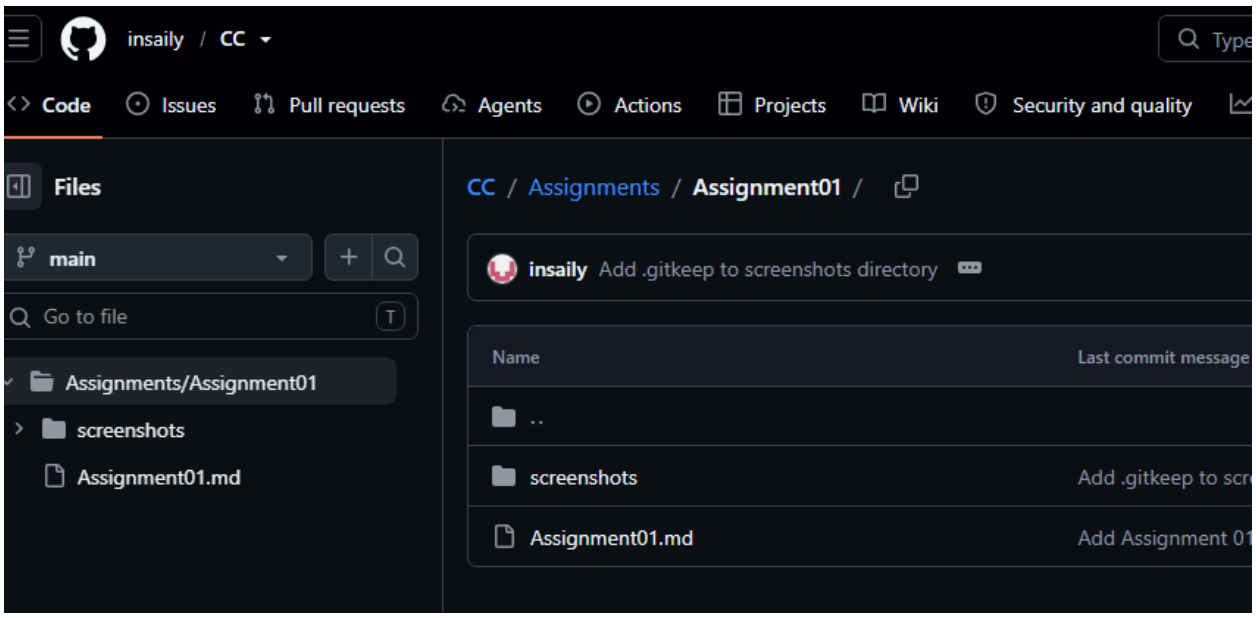
  gitea:
    image: gitea/gitea:1.22
    container_name: gitea
    restart: always
    depends_on:
      - db
    environment:
      USER_UID: 1000
      USER_GID: 1000
      GITEA__database__DB_TYPE: postgres
      GITEA__database__HOST: db:5432
      GITEA__database__NAME: gitea
      GITEA__database__USER: gitea
      GITEA__database__PASSWD: gitea_db_pass_123
      GITEA__server__ROOT_URL: http://40.80.88.34:3000/
    ports:
      - "3000:3000"
      - "2222:22"
    volumes:
      - gitea_data:/data

volumes:
  postgres_data:
  gitea_data:
insaily@ubuntu:~/gitea$
```

8. Confirm that both containers are running:

```
insaily@ubuntu:~/gitea$ sudo docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED          STATUS          PORTS
                                                  NAMES
1a33d9af2e9d   gitea/gitea:1.22   "/usr/bin/entrypoint…"   46 seconds ago   Up 45 seconds   0.0.0.0:3000->3000/tcp, [::]:3000-
>3000/tcp, 0.0.0.0:2222->22/tcp, [::]:2222->22/tcp   gitea
61bdfb03e4e8   postgres:15        "docker-entrypoint.s…"   51 seconds ago   Up 46 seconds   5432/tcp
                                                  gitea-db
insaily@ubuntu:~/gitea$
```

Task 2: Push the Same Repository to GitHub



```
insaily@ubuntu:~/CC-repo$ mkdir -p Assignments/Assignment01
insaily@ubuntu:~/CC-repo$ cd Assignments/Assignment01
insaily@ubuntu:~/CC-repo/Assignments/Assignment01$ dd if=/dev/zero of=file1.bin bs=1M count=101 status=progress
101+0 records in
101+0 records out
105906176 bytes (106 MB, 101 MiB) copied, 0.0471414 s, 2.2 GB/s
insaily@ubuntu:~/CC-repo/Assignments/Assignment01$ dd if=/dev/zero of=file2.bin bs=1M count=101 status=progress
101+0 records in
101+0 records out
105906176 bytes (106 MB, 101 MiB) copied, 0.0459038 s, 2.3 GB/s
insaily@ubuntu:~/CC-repo/Assignments/Assignment01$ dd if=/dev/zero of=file3.bin bs=1M count=101 status=progress
101+0 records in
101+0 records out
105906176 bytes (106 MB, 101 MiB) copied, 0.0473264 s, 2.2 GB/s
insaily@ubuntu:~/CC-repo/Assignments/Assignment01$ ls -lh *.bin
-rw-rw-r-- 1 insaily insaily 101M Oct 10 18:01 file1.bin
-rw-rw-r-- 1 insaily insaily 101M Oct 10 18:01 file2.bin
-rw-rw-r-- 1 insaily insaily 101M Oct 10 18:01 file3.bin
insaily@ubuntu:~/CC-repo/Assignments/Assignment01$ cd ~/CC-repo
insaily@ubuntu:~/CC-repo$ git add .gitattributes Assignments/Assignment01/file1.bin Assignments/Assignment01/file2.bi
n Assignments/Assignment01/file3.bin
insaily@ubuntu:~/CC-repo$ git commit -m "Add Git LFS large files"
[main 91dad99] Add Git LFS large files
 Committer: Ubuntu <insaily@ubuntu.vtvqdvf2kv4uxp3h3rnpurjf1d.rx.internal.cloudapp.net>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 4 files changed, 10 insertions(+)
 create mode 100644 .gitattributes
 create mode 100644 Assignments/Assignment01/file1.bin
 create mode 100644 Assignments/Assignment01/file2.bin
 create mode 100644 Assignments/Assignment01/file3.bin
insaily@ubuntu:~/CC-repo$ git lfs ls-files --size
f1753a034b * Assignments/Assignment01/file1.bin (106 MB)
f1753a034b * Assignments/Assignment01/file2.bin (106 MB)
f1753a034b * Assignments/Assignment01/file3.bin (106 MB)
insaily@ubuntu:~/CC-repo$
```

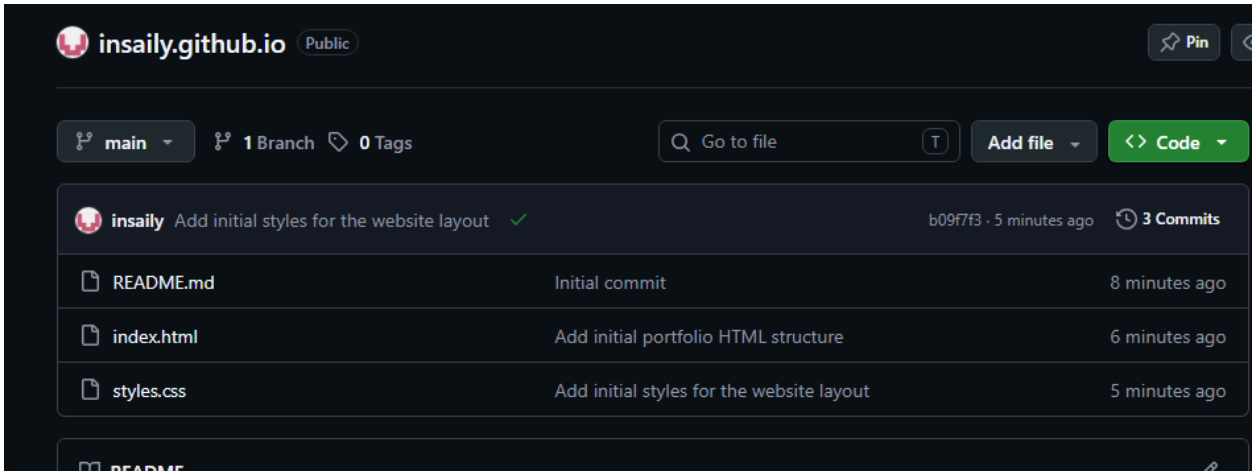
Task 3: Track Three Large Files with Git LFS

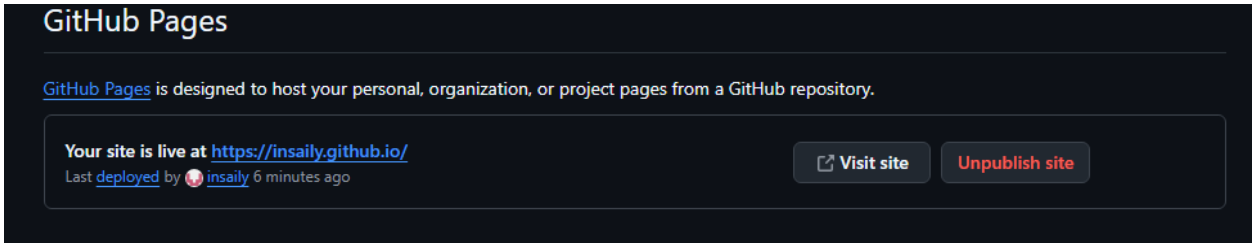
```
insaily@ubuntu:~$ git lfs install
Git LFS initialized.
insaily@ubuntu:~$ cd ~
insaily@ubuntu:~$ git clone https://github.com/insaily/CC.git
fatal: destination path 'CC' already exists and is not an empty directory.
insaily@ubuntu:~$ cd ~/CC
insaily@ubuntu:~/CC$ git lfs track "*.bin"
Not in a git repository.
insaily@ubuntu:~/CC$ cat .gitattributes
cat: .gitattributes: No such file or directory
insaily@ubuntu:~/CC$ ls -la ~/CC
total 12
drwxrwxr-x 3 insaily insaily 4096 Oct 10 17:39 .
drwxr-x--- 7 insaily insaily 4096 Oct 10 17:45 ..
drwxrwxr-x 3 insaily insaily 4096 Oct 10 17:39 Assignments
```

```
insaily@ubuntu: $ cd ~/CC-repo
insaily@ubuntu:~/CC-repo$ git lfs track "*.bin"
Tracking "*.bin"
insaily@ubuntu:~/CC-repo$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
insaily@ubuntu:~/CC-repo$ mkdir -p Assignments/Assignment01
insaily@ubuntu:~/CC-repo$ cd Assignments/Assignment01
insaily@ubuntu:~/CC-repo/Assignments/Assignment01$ dd if=/dev/zero of=file1.bin bs=1M count=101 status=progress
101+0 records in
101+0 records out
105906176 bytes (106 MB, 101 MiB) copied, 0.0471414 s, 2.2 GB/s
insaily@ubuntu:~/CC-repo/Assignments/Assignment01$ dd if=/dev/zero of=file2.bin bs=1M count=101 status=progress
101+0 records in
101+0 records out
105906176 bytes (106 MB, 101 MiB) copied, 0.0459038 s, 2.3 GB/s
insaily@ubuntu:~/CC-repo/Assignments/Assignment01$ dd if=/dev/zero of=file3.bin bs=1M count=101 status=progress
101+0 records in
101+0 records out
105906176 bytes (106 MB, 101 MiB) copied, 0.0473264 s, 2.2 GB/s
insaily@ubuntu:~/CC-repo/Assignments/Assignment01$ ls -lh *.bin
-rw-rw-r-- 1 insaily insaily 101M Oct 10 18:01 file1.bin
-rw-rw-r-- 1 insaily insaily 101M Oct 10 18:01 file2.bin
-rw-rw-r-- 1 insaily insaily 101M Oct 10 18:01 file3.bin
```

Task 4: Create a Portfolio or CV with GitHub Pages

Link: https://insaily.github.io/





```html
```

Insa Ilyas

Software Engineering Student | Aspiring Data Analyst

About Education Skills Projects

About Me

I am a Software Engineering student interested in Python, data analytics, artificial intelligence, and software development.

Activate Windows
Go to Settings to activate Windows.

Education

BS Software Engineering
Fatima Jinnah Women University

A-Levels
Punjab Group of Colleges (2021–2023)

Skills

- Python
- SQL
- C++ and OOP
- Data Analysis
- HTML and CSS

Activat
Go to Se

Projects

AI Safety and Accessibility Copilot

A project concept focused on identifying hazards and providing accessibility alerts.

Energy Consumption Analysis

A data analysis project exploring energy consumption patterns.

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